

SMALL COLLEGE DEPARTMENT NETWORK

Computer Networks – Micro Project / PBL-2

Cisco Packet Tracer Implementation Report

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Project Theme

A small, segmented and manageable college department network

AIM:

To design and implement a small but realistic college department network in Cisco Packet Tracer. The network separates Administration, Students/Lab and Faculty into independent VLANs, uses a Cisco router for inter-VLAN communication, and uses DHCP for automatic IPv4 configuration. The design also provides a foundation for centralized DNS/HTTP services and basic access-control policies.

PROJECT OVERVIEW

The project models the network of a single college department rather than an entire campus. This keeps the topology compact while demonstrating concepts used in real organizational networks. Two Administration PCs, four Student/Lab PCs and two Faculty PCs are connected to a Cisco 2960 switch. A Cisco 2911 router provides the Layer-3 gateways, and a server is connected to the Faculty/Server VLAN.

1. PROBLEM STATEMENT AND OBJECTIVES

A college department normally contains users with different responsibilities and different levels of access. Placing every computer in one flat LAN makes the network harder to manage and provides weak logical separation. The proposed solution divides the department into VLANs and routes between them through a router.

1.1 Objectives

- Create separate VLANs for Administration, Students/Lab and Faculty.
- Use an 802.1Q trunk between the switch and router.
- Configure Router-on-a-Stick for communication between VLANs.
- Use DHCP to automatically assign addresses to client PCs.
- Use a static address for the central server.
- Provide DNS and HTTP services as the department's centralized network services.
- Apply a basic ACL policy so student traffic can be restricted from sensitive departmental devices.
- Verify the implementation using Cisco IOS and PC diagnostic commands.

1.2 Real-World Relevance

The same principles are used in offices, schools, hospitals and other organizations. VLANs reduce broadcast-domain size and provide logical separation; routing enables controlled communication between those networks; DHCP reduces manual configuration; DNS makes services easier to locate; and ACLs can enforce basic security policies.

2. REQUIREMENTS

2.1 Packet Tracer Devices

Device	Qty.	Role
Cisco 2911 Router	1	Inter-VLAN routing and DHCP
Cisco 2960 Switch	1	VLANs, access ports and trunk
PC-PT	8	Department end devices
Server-PT	1	Central server for network services
Copper Straight-Through	10	Device connections

2.2 Network Requirements

- Three IPv4 /24 networks.
- Three VLANs with separate broadcast domains.
- One router-to-switch trunk.
- DHCP pools for all three client VLANs.
- A static IP for the server.
- DNS and HTTP services for the department portal.
- Basic extended ACL rules for student traffic.

3. NETWORK TOPOLOGY

The physical topology consists of one Cisco 2911 router, one Cisco 2960 switch, eight PCs and one server. The switch is the central LAN device. The router is connected to the switch through a trunk link and uses subinterfaces to represent the three VLAN gateways.

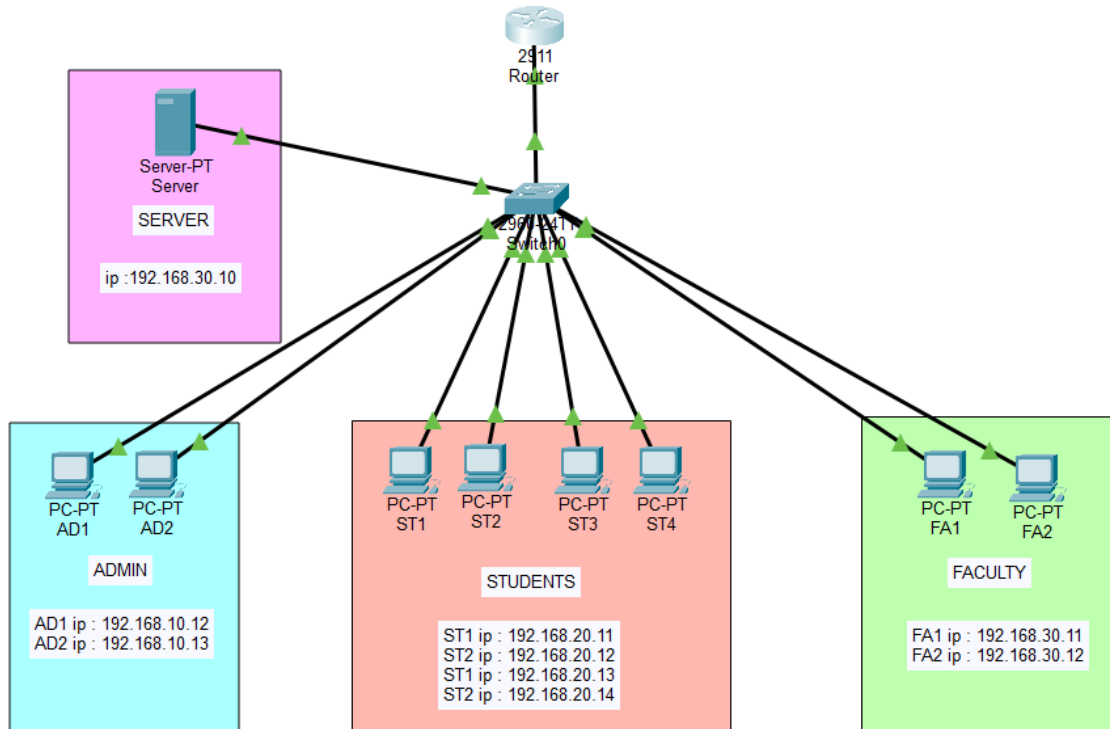


Figure 1: Final Packet Tracer topology supplied for the project.

3.1 Logical Grouping

Group	Devices	VLAN	Subnet
Administration	AD1, AD2	10	192.168.10.0/24
Students / Lab	ST1, ST2, ST3, ST4	20	192.168.20.0/24
Faculty	FA1, FA2	30	192.168.30.0/24
Server	Server	30	192.168.30.0/24

3.2 Port Allocation

Switch Port	Device	Configuration
Fa0/1	Router G0/0	Trunk
Fa0/2–Fa0/3	AD1, AD2	Access VLAN 10
Fa0/4–Fa0/7	ST1–ST4	Access VLAN 20
Fa0/8–Fa0/9	FA1, FA2	Access VLAN 30
Fa0/10	Server	Access VLAN 30

The remaining switch ports are unused in this model. The trunk port is intentionally not listed as an access member in the VLAN table because it carries tagged traffic for multiple VLANs.

4. VLAN CONFIGURATION

VLANs logically divide the switch into separate Layer-2 broadcast domains. In this project, VLAN 10 represents Administration, VLAN 20 represents Students/Lab, and VLAN 30 represents Faculty and the central server.

VLAN ID	Name	Network	Gateway	Purpose
10	ADMIN	192.168.10.0/24	192.168.10.1	Administrative users
20	STUDENTS	192.168.20.0/24	192.168.20.1	Student/Lab users
30	FACULTY	192.168.30.0/24	192.168.30.1	Faculty and server

4.1 Verified Switch Output

The following screenshot is the actual switch output from the implemented Packet Tracer model. It confirms that the required access ports are assigned to the intended VLANs.

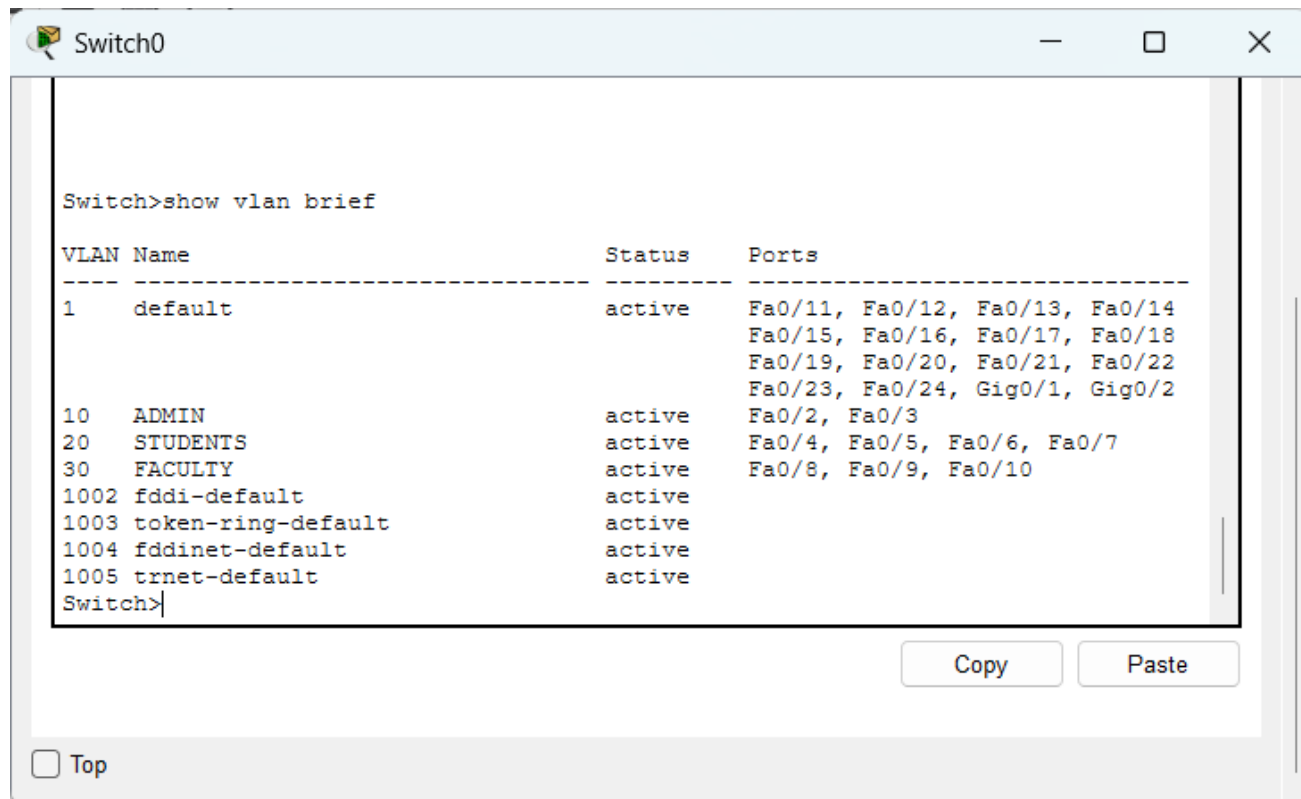


Figure 2: Switch output – show vlan brief.

4.2 Interpretation of the Output

- Fa0/2 and Fa0/3 are assigned to VLAN 10 (ADMIN).
- Fa0/4 through Fa0/7 are assigned to VLAN 20 (STUDENTS).
- Fa0/8 through Fa0/10 are assigned to VLAN 30 (FACULTY/server).
- Fa0/1 is the router connection and is configured as the trunk; therefore it is not shown as an access port under VLAN 10, 20 or 30.

5. IP ADDRESSING AND INTER-VLAN ROUTING

Each VLAN uses a separate /24 IPv4 network. The router provides the first usable address in each subnet as the default gateway. This addressing plan makes the network easy to understand and expand.

5.1 IP Addressing Plan

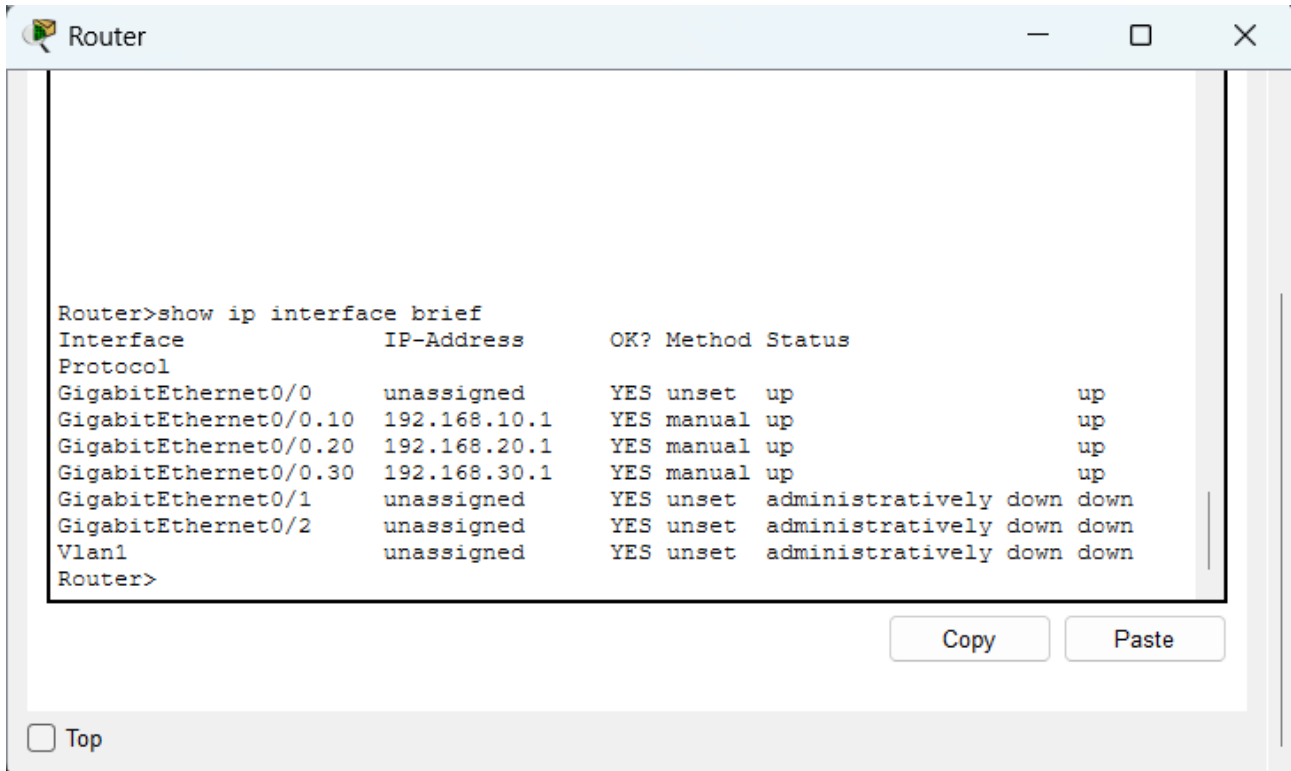
Device / VLAN	IP Address	Subnet Mask	Default Gateway	Method
ADMIN VLAN 10	192.168.10.0/24	255.255.255.0	192.168.10.1	DHCP clients
STUDENTS VLAN 20	192.168.20.0/24	255.255.255.0	192.168.20.1	DHCP clients
FACULTY VLAN 30	192.168.30.0/24	255.255.255.0	192.168.30.1	DHCP clients
Server	192.168.30.10	255.255.255.0	192.168.30.1	Static

5.2 Router Subinterfaces

Interface	VLAN	IP Address	Function
G0/0.10	10	192.168.10.1	Administration gateway
G0/0.20	20	192.168.20.1	Student gateway
G0/0.30	30	192.168.30.1	Faculty/server gateway

Router-on-a-Stick is used because a single physical router interface can carry multiple VLANs when the switch-to-router link is a trunk. The router subinterfaces use IEEE 802.1Q VLAN tags to identify the incoming traffic.

5.3 Verified Router Output



```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0       unassigned      YES unset  up
GigabitEthernet0/0.10    192.168.10.1    YES manual  up
GigabitEthernet0/0.20    192.168.20.1    YES manual  up
GigabitEthernet0/0.30    192.168.30.1    YES manual  up
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router>
```

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Figure 3: Router output – show ip interface brief.

The output confirms that G0/0 and all three subinterfaces are operational. The configured addresses 192.168.10.1, 192.168.20.1 and 192.168.30.1 correspond to the three VLAN gateways.

6. DHCP CONFIGURATION

Dynamic Host Configuration Protocol (DHCP) is configured on the Cisco 2911 router. It automatically provides client PCs with an IPv4 address, subnet mask, default gateway and DNS server. Separate DHCP pools are used because each VLAN belongs to a different IP subnet.

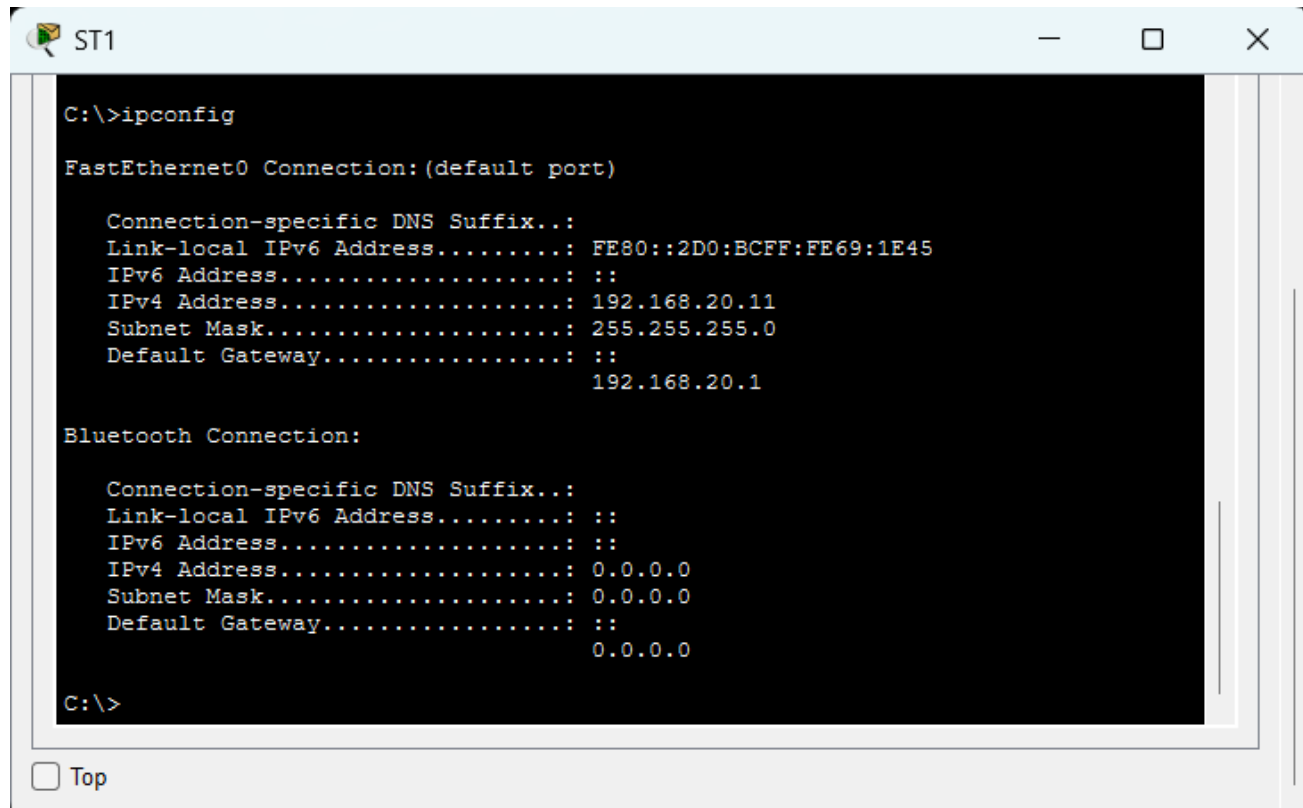
6.1 DHCP Pools

Pool	Network	Default Gateway	DNS Server	Excluded Range
ADMIN	192.168.10.0/24	192.168.10.1	192.168.30.10	192.168.10.1–10
STUDENTS	192.168.20.0/24	192.168.20.1	192.168.30.10	192.168.20.1–10
FACULTY	192.168.30.0/24	192.168.30.1	192.168.30.10	192.168.30.1–10

The first ten addresses of each subnet are excluded so that gateway and infrastructure addresses remain reserved. Client addresses are then leased dynamically from the remaining address range.

6.2 DHCP Client Verification

The Student PC shown below received 192.168.20.11, confirming that the PC is receiving an address from the Student VLAN's DHCP scope rather than being manually assigned an address.



```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:BCFF:FE69:1E45
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.20.11
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.20.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

Figure 4: Student PC – ipconfig output showing DHCP-assigned IP information.

6.3 DHCP Lease Verification on Router

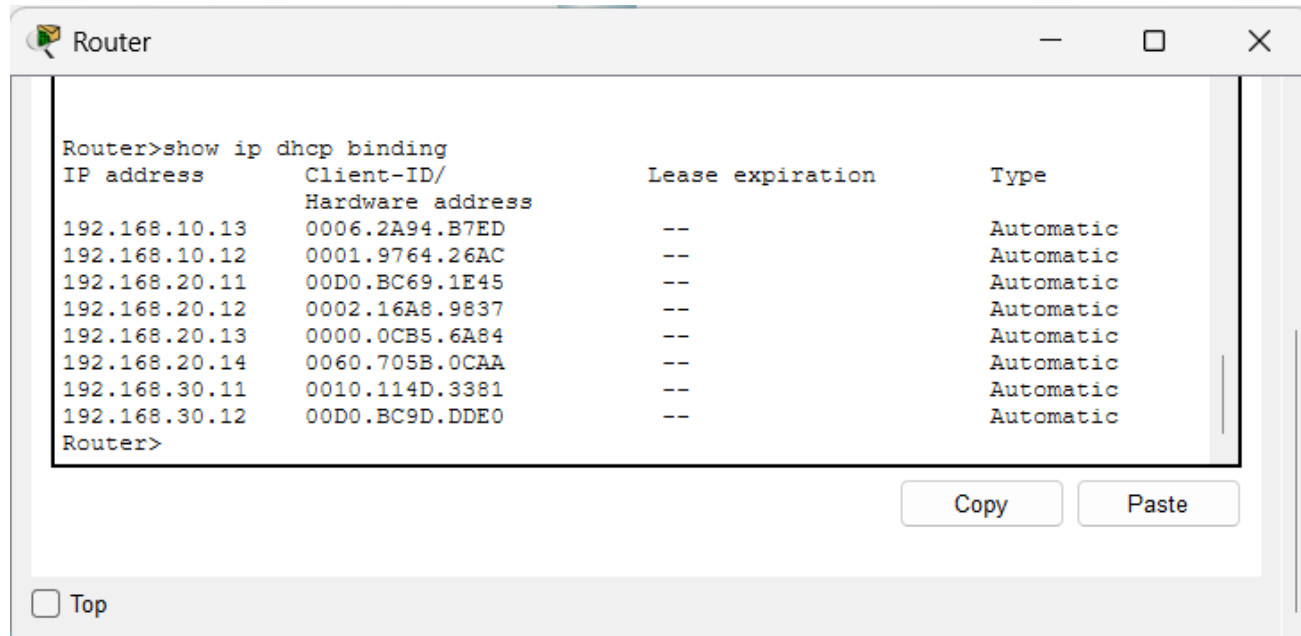


Figure 5: Router output – show ip dhcp binding.

7. CLIENT IP ADDRESS ALLOCATION

The DHCP binding output and the final topology show the following client addresses. The exact order of DHCP allocation depends on the sequence in which the clients requested their leases.

Client	Department	IP Address	Gateway	Addressing
AD1 / PC0	Administration	192.168.10.12	192.168.10.1	DHCP
AD2 / PC1	Administration	192.168.10.13	192.168.10.1	DHCP
ST1 / PC2	Students	192.168.20.11	192.168.20.1	DHCP
ST2 / PC3	Students	192.168.20.12	192.168.20.1	DHCP
ST3 / PC4	Students	192.168.20.13	192.168.20.1	DHCP
ST4 / PC5	Students	192.168.20.14	192.168.20.1	DHCP
FA1 / PC7	Faculty	192.168.30.11	192.168.30.1	DHCP
FA2 / PC6	Faculty	192.168.30.12	192.168.30.1	DHCP
Server	Faculty/Server	192.168.30.10	192.168.30.1	Static

7.1 What the DHCP Binding Table Proves

- 192.168.10.12 and 192.168.10.13 were leased to the two Administration PCs.
- 192.168.20.11 through 192.168.20.14 were leased to the four Student PCs.
- 192.168.30.11 and 192.168.30.12 were leased to the two Faculty PCs.
- All shown client leases are marked Automatic, confirming DHCP allocation.
- The server address 192.168.30.10 is outside the dynamic client range and is reserved as a static infrastructure address.

8. BASIC CONNECTIVITY TESTING

After VLANs, routing and DHCP are configured, connectivity can be verified from the PCs using the ping and ipconfig commands.

Test	Expected Result	Purpose
ipconfig on Admin PC	192.168.10.x	Verify DHCP and VLAN 10
ipconfig on Student PC	192.168.20.x	Verify DHCP and VLAN 20
ipconfig on Faculty PC	192.168.30.x	Verify DHCP and VLAN 30
ping 192.168.20.1 from ST1	Reply	Verify Student gateway
ping 192.168.10.1 from AD1	Reply	Verify Admin gateway
ping 192.168.30.1 from FA1	Reply	Verify Faculty gateway

Successful gateway pings demonstrate that the client can reach the router interface serving its VLAN. Inter-VLAN tests can then be performed to verify Router-on-a-Stick operation.

9. CENTRAL SERVER, DNS AND WEB SERVICE DESIGN

The project reserves Server-PT at 192.168.30.10 as the central service host. Keeping a server on a stable static address ensures that clients can consistently locate the services it provides.

9.1 Server IP Configuration

Parameter	Value
IP Address	192.168.30.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1
DNS Server	192.168.30.10
VLAN	30

9.2 DNS Service

The intended DNS configuration maps a human-readable local hostname to the server. A suitable record for the project is:

DNS Name	IP Address	Purpose
portal.college.local	192.168.30.10	Department portal

Client PCs receive 192.168.30.10 as their DNS server through DHCP. When DNS is enabled on the server and the record is added, users can access the portal by name instead of remembering the numeric address.

9.3 HTTP Service

The Server-PT can host a simple internal department webpage using the HTTP service. Suggested content includes department notices, student information, faculty information and links to internal resources.

Example URL: <http://portal.college.local>

10. BASIC ACL SECURITY DESIGN

A basic extended ACL can be applied to the Student VLAN interface to prevent student devices from directly reaching sensitive Administration and Faculty devices. The server remains accessible because students need the department portal.

Traffic	Policy	Reason
Students → 192.168.30.10	PERMIT	Allow access to department server
Students → 192.168.10.0/24	DENY	Protect Administration devices
Students → Faculty devices	DENY	Protect Faculty devices
Other required traffic	PERMIT	Maintain normal network operation

This security design is an extension of the demonstrated VLAN/routing/DHCP model. It illustrates how the project can move from simple connectivity to controlled organizational access.

11. IMPLEMENTATION SUMMARY

Component	Status / Evidence	Key Result
VLANs	Verified in Figure 2	VLAN 10, 20 and 30 correctly assigned
Router gateways	Verified in Figure 3	G0/0.10, .20 and .30 are up/up
DHCP client IP	Verified in Figure 4	ST1 received 192.168.20.11
DHCP leases	Verified in Figure 5	All eight client leases visible
Server IP	Shown in final topology	192.168.30.10
DNS	Design specified	portal.college.local → 192.168.30.10
HTTP	Design specified	Internal department portal
ACL	Security design specified	Student access restrictions

12. TROUBLESHOOTING AND VERIFICATION COMMANDS

Device	Command / Action	What It Checks
Switch	show vlan brief	VLAN names and access-port membership
Switch	show interfaces trunk	Trunk status and VLAN transport
Router	show ip interface brief	Interface/subinterface status and IPs
Router	show ip dhcp pool	DHCP pool configuration
Router	show ip dhcp binding	Active DHCP leases
PC	ipconfig	Assigned IP, mask, gateway and DNS
PC	ping <address>	Basic IP connectivity

13. DEMONSTRATION PLAN

1. Start with the topology and explain the three user groups.
2. Open the switch CLI and show show vlan brief.
3. Open the router CLI and show show ip interface brief.
4. Open ST1 and show ipconfig to prove DHCP assignment.
5. Open the router and show show ip dhcp binding to prove that the router has issued leases.
6. Explain the IP addressing scheme and the role of each default gateway.
7. Demonstrate inter-VLAN connectivity with ping.
8. If DNS/HTTP/ACL are enabled, demonstrate portal.college.local and the student access restrictions.

CONCLUSION:

The Small College Department Network provides a compact and realistic example of an organizational computer network. The implemented model successfully demonstrates VLAN-based departmental separation, router subinterfaces for inter-VLAN routing, and automatic client addressing through DHCP. The supplied Packet Tracer evidence confirms the VLAN membership, router gateway configuration and DHCP leases. The architecture also provides a practical base for centralized DNS/HTTP services and ACL-based access control. Overall, the project combines multiple core Computer Networking concepts into a single working model that is sufficiently complex for a micro-project while remaining understandable and manageable.